

EXP : 5 ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS

OBJECTIVES:

To write an Embedded C program for Manual Electronic counter using Arduino IDE and Proteus.

CODE:

```
int x0,x1,x2,x3,x4,x5,x6,x7,x8,x9;

int delay_time=200;

void setup() {
    //configure pin2 as an input and enable the internal pull-up resistor
    pinMode(12, INPUT_PULLUP);
    pinMode(1,OUTPUT); pinMode(2,OUTPUT);
    pinMode(3,OUTPUT);
    pinMode(4,OUTPUT);
    pinMode(5,OUTPUT);
    pinMode(6,OUTPUT);
    pinMode(7,OUTPUT);
}

void loop() {
    //read the pushbutton value into a variable
    int sensorVal = digitalRead(12);
    if (sensorVal == LOW) {
        x0=true;
    }

    while(x0){
```

```
zero();  
sensorVal = digitalRead(12);  
if (sensorVal == LOW) {  
    x1=true;  
    x0=false;  
}  
}
```

```
while(x1){  
    one();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
        x2=true;  
        x1=false;  
    }  
}
```

```
while(x2){  
    two();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
        x3=true;  
        x2=false;  
    }  
}
```

```
while(x3){  
    three();  
    sensorVal = digitalRead(12);
```

```
if (sensorVal == LOW) {  
    x4=true;  
    x3=false;  
}  
}
```

```
while(x4){  
    four();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
        x5=true;  
        x4=false;  
    }  
}
```

```
while(x5){  
    five();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
        x6=true;  
        x5=false;  
    }  
}
```

```
while(x6){  
    six();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
        x7=true;
```

```
    x6=false;
}
}
```

```
while(x7){
    seven();
    sensorVal = digitalRead(12);
    if (sensorVal == LOW) {
        x8=true;
        x7=false;
    }
}
```

```
while(x8){
    eight();
    sensorVal = digitalRead(12);
    if (sensorVal == LOW) {
        x9=true;
        x8=false;
    }
}
```

```
while(x9){
    nine();
    sensorVal = digitalRead(12);
    if (sensorVal == LOW) {
        x0=true;
        x9=false;
    }
}
```

```
}  
}
```

```
void zero()  
{  
  for(int i=1;i<7;i++)  
  {  
    digitalWrite(i,HIGH);  
    digitalWrite(7,LOW);  
  }  
  delay(delay_time);  
}
```

```
void one()  
{  
  digitalWrite(1,LOW);  
  digitalWrite(2,HIGH);  
  digitalWrite(3,HIGH);  
  digitalWrite(4,LOW);  
  digitalWrite(5,LOW);  
  digitalWrite(6,LOW);  
  digitalWrite(7,LOW);  
  delay(delay_time);  
}
```

```
void two()  
{  
  digitalWrite(1,HIGH);  
  digitalWrite(2,HIGH);
```

```
digitalWrite(3,LOW);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,LOW);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}
```

```
void three()  
{  
    digitalWrite(1,HIGH);  
    digitalWrite(2,HIGH);  
    digitalWrite(3,HIGH);  
    digitalWrite(4,HIGH);  
    digitalWrite(5,LOW);  
    digitalWrite(6,LOW);  
    digitalWrite(7,HIGH);  
    delay(delay_time);  
}
```

```
void four()  
{  
    digitalWrite(1,LOW);  
    digitalWrite(2,HIGH);  
    digitalWrite(3,HIGH);  
    digitalWrite(4,LOW);  
    digitalWrite(5,LOW);  
    digitalWrite(6,HIGH);  
    digitalWrite(7,HIGH);  
}
```

```
    delay(delay_time);  
}
```

```
void five()  
{  
    digitalWrite(1,HIGH);  
    digitalWrite(2,LOW);  
    digitalWrite(3,HIGH);  
    digitalWrite(4,HIGH);  
    digitalWrite(5,LOW);  
    digitalWrite(6,HIGH);  
    digitalWrite(7,HIGH);  
    delay(delay_time);  
}
```

```
void six()  
{  
    digitalWrite(1,HIGH);  
    digitalWrite(2,LOW);  
    digitalWrite(3,HIGH);  
    digitalWrite(4,HIGH);  
    digitalWrite(5,HIGH);  
    digitalWrite(6,HIGH);  
    digitalWrite(7,HIGH);  
    delay(delay_time);  
}
```

```
void seven()  
{
```

```
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,LOW);  
digitalWrite(5,LOW);  
digitalWrite(6,LOW);  
digitalWrite(7,LOW);  
delay(delay_time);  
}
```

```
void eight()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}
```

```
void nine()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,HIGH);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,LOW);
```

```
digitalWrite(6,HIGH);
```

```
digitalWrite(7,HIGH);
```

```
delay(delay_time);
```

```
}
```

OUTPUT:

